

URS PLPI BAR Automation
PLPI/BAR/URS/01/v1

Author _____ Date _____
Yogeshkumar Vimalan
Business Analyst

Reviewer _____ Date _____
Amit Sanandiya
Project Manager

Reviewer _____ Date _____
Juston Rodrigues
Team Leader

Reviewer _____ Date _____
Anthony Fernandes
Team Leader

Approver _____ Date _____
Beauty Dadhaniya
QA

URS PLPI BAR AutomationPLPI/BAR/URS/01/v1

Revision History

Version	Previous version	Reason for revision	Issued
1	NA	New URS prepared for PLPI Batch Record Automation covering B&S Batch Add, Printing modules and Leaflet Folding.	Aug 2026

Contents

Revision History	2
1. Introduction	4
2. Scope	4
2.1 Out of Scope	
3. Abbreviations	4
4. User Requirement and Specification	5
4.1 B&S Batch Add and BAR Creation	
4.2 Printing Module	
4.3 Leaflet Folding	
5. User Access	7
6. Testing	8
7. Documents	8
8. Support	8

URS PLPI BAR Automation

PLPI/BAR/URS/01/v1

1. Introduction

The PLPI Batch Record Automation Phase 2 project is a phased digital improvement initiative intended to replace paper-dependent activities with controlled electronic workflow records while retaining established GMP checks. This phase begins when an approved batch reaches B&S Batch Add and continues through BAR generation, printing activities and leaflet folding.

The change introduces controlled work queues, electronic BAR records, route-based printing actions, quantity and evidence capture, user sign-off, audit history and status-controlled handoffs. Functional screen behaviour and technical implementation will be defined in the Functional Specification and Design Specification after this URS is approved.

2. Scope

The scope covers B&S Batch Add and BAR generation, BAR verification and digital line clearance, the shared Printer module, Label Printing, Leaflet Printing, Carton Issuing, Braille Printing where required, and Leaflet Folding. It includes queue eligibility, search and record display, approved document and artwork access, quantity controls, test or master review, evidence capture, comments and reasons, electronic sign-off, audit trail, exception handling and downstream handoff.

2.1 Out of Scope

Pre-Assembly, Production Control, assembly-room activities, Post-Assembly QC, Pre-QP and QP Approval are outside scope except for the controlled handoff from Leaflet Folding.

3. Abbreviations

BAR	Batch Assembly Record
GMP	Good Manufacturing Practice
PCL	Product Check Log
PLPI	PLPI business application and controlled workflow
QP	Qualified Person
RP / RPi	Responsible Person / Responsible Person import
URS	User Requirement Specification

URS PLPI BAR Automation

PLPI/BAR/URS/01/v1

4. User Requirement and Specification

The following user requirements define the controls required to implement B&S Batch Add, Printing modules and Leaflet Folding within PLPI.

4.1 B&S Batch Add and BAR Creation

URS ID	Requirements
4.1.2	The system shall allow the user to search and select eligible records using B&S batch number, product, country, site, status and other available batch criteria.
4.1.3	The system shall display the product, pack, reference, expiry, quantity, order, location and route information required to confirm the selected batch.
4.1.4	The system shall permit compatible source records to be combined only when the configured product, batch and expiry criteria agree, and shall retain traceability to every source record.
4.1.5	The system shall require confirmation of the selected B&S batch before creating one controlled electronic BAR.
4.1.6	The system shall provide BAR verification and digital line-clearance checks for batch identity, product details, approved source records and mandatory documentation.
4.1.7	BAR sign-off shall capture the authenticated user, role, date and time, lock the completed checks and route the batch to the Printing module.

4.2 Printing Module

URS ID	Requirements
4.2.1	The system shall provide one controlled Printing module containing separate work queues for Label Printing, Leaflet Printing, Carton Issuing and Braille Printing, with access restricted by the user's authorised role.
4.2.2	The Label Printing queue shall receive batches after BAR issue. The Leaflet Printing queue shall receive batches after Label Printing. Carton Issuing shall list only reboxing batches after Leaflet Printing, and Braille Printing shall list only relabelling batches for which braille is required after Leaflet Printing. Each queue shall support search by B&S batch number and manufacturing lot number.

URS PLPI BAR Automation

PLPI/BAR/URS/01/v1

URS ID	Requirements
4.2.3	For the selected batch, the system shall display the product, strength, pack size, B&S batch number, manufacturing lot number, expiry, process route, component description and reference, size, required quantity, stock or in-hand quantity, location and current printing status needed to complete the activity.
4.2.4	The Printing module shall provide controlled access to the populated BAR and the approved artwork and supporting records applicable to the route, including label, carton, peel, braille, mock-up, continuation-page and leaflet documents. The system shall retain the applicable document reference or version and an audit record of controlled document access.
4.2.5	Label Printing shall derive the required label lines from the approved batch route. For reboxing batches, printing shall remain blocked until the Received As and Assembled As information is complete and signed. A test label shall be accepted before the full run, and each label line shall show its reference, size, required quantity, location, in-hand quantity, print action, completion status and completed-by/date information.
4.2.6	For every printable line, the system shall distinguish the approved required quantity from the quantity already printed and the quantity remaining. It shall support controlled partial printing and additional printing; any quantity above the approved requirement shall be identified as extra and shall require the user to select or enter a reason before printing.
4.2.7	The system shall save printing progress by batch and component line, retain each print run with quantity, printer, user, date and time, and prevent Print Done until every applicable line is complete. Line completion and final completion shall be attributable to the authenticated user and retained in the batch audit history.
4.2.8	Leaflet Printing shall display each approved leaflet reference, size, quantity and location, open the controlled leaflet PDF for printing, require the master or test copy to be reviewed before the full run, and enable Mark as Done only after the applicable leaflet has been printed. Extra leaflet quantities shall require a recorded reason.

URS PLPI BAR Automation

PLPI/BAR/URS/01/v1

URS ID	Requirements
4.2.9	Carton Issuing shall be available only for reboxing batches and shall record carton reference, required quantity, on-hand quantity, issue location, extra quantity and reason where applicable, and the confirming user and date/time. Braille Printing shall be available only for relabelling batches requiring braille and shall apply the same line-level quantity, extra-print, completion and audit controls used for controlled label printing.
4.2.10	When a printing activity is completed, the system shall update the batch status and route it to the next applicable activity: Label Printing to Leaflet Printing where required; Leaflet Printing to Carton Issuing for reboxing or to Braille Printing where required for relabelling; and completed route-specific printing to Leaflet Folding or the next configured stage. Completed records shall remain available for controlled review and audit.

4.3 Leaflet Folding

URS ID	Requirements
4.3.1	The Leaflet Folding module shall display only batches that have completed all printing and route-specific preparation required before folding.
4.3.2	The folding queue shall support search by B&S batch number and manufacturing lot number and shall display the required product, batch, leaflet, quantity and status information.
4.3.3	Opening a folding record shall provide controlled access to the applicable BAR, approved leaflet reference and supporting batch documents.
4.3.4	The system shall require the operator to confirm that the selected batch and leaflet agree with the approved BAR before completion.
4.3.5	The system shall record the folded quantity and shall prevent completion where the leaflet reference, fold format, quantity or batch identity is missing or incorrect.
4.3.6	Leaflet Folding completion shall capture the authenticated user, role, date and time and retain the completed record in batch history.

5. User Access

User access shall be role based. B&S Batch Add users shall select eligible source records, generate the BAR, complete the required checks and sign off the release to printing. Printing users shall access only the printing options and batches

URS PLPI BAR Automation

PLPI/BAR/URS/01/v1

appropriate to their responsibilities. Leaflet Folding users shall complete assigned folding activities. QA/RP, Operations and authorised IT users shall have review, exception, reporting or administration access according to approved responsibility.

The system shall uniquely identify each user, prevent shared-user attribution and restrict administration, master-data maintenance, correction, reprint and exception-release actions to authorised roles. Access changes and privileged activities shall be auditable.

6. Testing

The PLPI system owner shall coordinate verification testing with IT, QA and Operations. Testing shall trace to the approved URS requirements and shall cover role access; queue eligibility and searching; BAR generation and duplicate prevention; line-clearance controls; route-based label, leaflet, carton and braille activities; test and master review; partial and extra quantities; evidence and reasons; user sign-off; status handoff; audit history; holds, corrections and reprints; record locking; technical failure behaviour; and Leaflet Folding completion.

Regression testing shall confirm that the upstream Batch Checker handoff, downstream pre-assembly receipt and unchanged PLPI functions continue to operate as approved. Representative operational and quality users shall complete user acceptance testing before release.

7. Documents

Applicable SOPs, work instructions, forms, training material, validation records and controlled project documents shall be created or updated to describe BAR generation, verification, printing queues, quantity and evidence controls, route handling, Leaflet Folding, exception management, reprints, record review and electronic sign-off. Superseded paper activities shall be retired through the approved change-control and document-control process.

8. Support

IT shall provide controlled support for PLPI configuration, access, printer and device connectivity, document and artwork links, workflow queues, audit records, backup, recovery and incident resolution. Business, QA/RP and Operations stakeholders shall support controlled process use, master-data accuracy, exception decisions and review of future changes.